Connecting a buzzer directly to a Raspberry Pi 5 GPIO pin causes continuous beeping because the Pi's pins operate at 3.3V and can only safely supply about 16mA of current. Direct connections either pull the pin into a floating state or draw too much current, which risks permanently burning out the Raspberry Pi's processor.
To safely switch a 5V buzzer using a 3.3V Raspberry Pi signal, you must use an **NPN Transistor** acting as a digital switch, paired with a **Current-Limiting Resistor**.

**Required Components**

- **1x NPN Transistor (2N2222 or BC547):** Acts as the heavy-duty switch to route 5V power to the buzzer without touching the Pi's delicate logic pins.
- **1x $1\text{ k}\Omega$ Resistor:** Protects the Raspberry Pi GPIO pin by limiting the current flowing into the transistor base.
- **1x 5V Active Buzzer:** Generates sound automatically when 5V DC is applied.
- *(Optional)* **1x 1N4007 Diode:** Wired across the buzzer to absorb reverse-voltage spikes (highly recommended if the buzzer is magnetic rather than piezoelectric).

**Raspberry Pi 5 Buzzer Wiring Schematic**

Plaintext

```
============================================================================
RASPBERRY PI 5 SAFE BUZZER DRIVER CIRCUIT
============================================================================

 [ RPi 5V Pin (Pin 2 or 4) ] ────────────────────────────> [ Buzzer (+) Positive ]
                                  │
                                  │
                          [ Buzzer (-) Negative ]
                                  │
                                  │
                             (Collector)
 [ RPi GPIO 17 (Pin 11) ] ─────[ 1kΩ Resistor ]──────────> (Base)  2N2222 NPN
Transistor
                             (Emitter)
                                  │
                                  │
 [ RPi GND (Pin 6 or 9) ] ──────────────────────────────────────────┘
============================================================================
```

**How it Works:**
When the Raspberry Pi sends a HIGH (3.3V) signal from GPIO 17, it passes safely through the $1\text{ k}\Omega$ resistor and turns the transistor "ON". The transistor then completes the circuit to Ground, allowing the 5V power from the Pi to flow through the buzzer and make it scream.

**Python Control Script (gpiozero)**

Since you are using a Raspberry Pi 5, the modern standard for GPIO control is the gpiozero library. This script handles both the single door beep and the continuous emergency siren.

Python
```python
from gpiozero import Buzzer
from time import sleep

# Initialize the buzzer on GPIO 17
alarm_buzzer = Buzzer(17)

def door_beep():
    """Single 100ms beep for door open/close events."""
    print("[EVENT] Main Door Toggled")
    alarm_buzzer.on()
    sleep(0.1)
    alarm_buzzer.off()

def emergency_alarm():
    """Continuous pulsing alarm for Fire/Earthquake."""
    print("🚨 [ALARM] EMERGENCY DETECTED! 🚨")
    try:
        while True:
            alarm_buzzer.on()
            sleep(0.5)
            alarm_buzzer.off()
            sleep(0.5)
    except KeyboardInterrupt:
        alarm_buzzer.off()
        print("Alarm silenced.")

# --- Test the functions ---
if __name__ == '__main__':
    # Test door beep
    door_beep()
```

```
sleep(2)

# Trigger emergency alarm
emergency_alarm()
```